

Newton, *Principia*: Definitions, Laws, and Book I, Sections I-II

The 1687 text in modern mathematical language, in Newton's order, with the
changes of 1713 and 1726

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About this rendering

This is a rendering of the first edition (1687) of Newton's *Philosophiæ Naturalis Principia Mathematica*, made from the Newton Project TEI transcriptions kept in this repository: the Definitions and their Scholium (docs/m1/NATP00075.xml), the Axioms or Laws of Motion (NATP00076.xml) and Book I (NATP00077.xml). It covers the Definitions, the Laws with their Corollaries and Scholium, Book I Section I (Lemmas I-XI and the Scholium), and Book I Section II through Proposition IV and its Scholium. Propositions V-X, which close Section II, are not included.

Every item keeps Newton's number, order and concepts. Statements are put into modern mathematical language; proofs follow Newton's steps and add none. Where a modern formula helps, it follows Newton's statement under *In modern terms*. Newton's figures are not reproduced; the letters refer to them, and the descriptions are enough to redraw them. Square brackets mark the few notes added for a modern reader. Long physical or historical explanations are condensed, in Newton's order.

The changes of the second edition (1713) and the third (1726) follow each item, marked **1713** and **1726**. They come from the Newton Project TEI of the 1713 Definitions, Laws and Book I (NATP00080–NATP00082, in this repository) and of the 1726 Definitions, Laws and Book I (NATP00085–NATP00087, in docs/m4). The three editions were aligned paragraph by paragraph; changes of spelling, punctuation and abbreviation are omitted, and only changes of content or of mathematical language are given. A summary of the pattern follows.

Vocabulary.

Newton	Modern meaning
ratio of equality; <i>ultimately equal</i>	$X/Y \rightarrow 1$ in the stated limit
ultimate ratio of vanishing quantities	$\lim X/Y$ as $X, Y \rightarrow 0$
first ratio of nascent quantities	$\lim X/Y$ as X, Y start from 0
duplicate, triplicate ratio	ratio of squares, of cubes
subduplicate, sesquiplicate ratio	ratio of square roots, of 3/2 powers
<i>A applied to B</i>	A/B
<i>componendo</i>	summing proportional terms
<i>in consequentia</i>	in the direction of motion

Bold letters are vectors; $\times$ is the planar cross product $\mathbf{a} \times \mathbf{b} = a_1 b_2 - a_2 b_1$.

How the editions differ

1713, the limiting arguments are tightened. Lemma I now says that the quantities actually come nearer than any given difference before the end of the finite time. Lemma VI gets a new proof from the continuity of the curvature. The proofs of Lemmas VII–IX enlarge the vanishing figure so that its lines stay finite while the original shrinks.

1713, regularity premises enter the statements. Lemma X, which assumed a “regular” force, now assumes a finite force, either constant or continually increasing or decreasing. Lemma XI is restricted to curves with finite curvature at the point of contact.

1713, the sagitta becomes the measure of force. Lemma XI gains corollaries on sagittas; Proposition I gains six corollaries measuring force by the sagitta of arcs described in equal times; Proposition IV is proved anew through the versed sines of those arcs, and its Scholium compares centripetal force with gravity through the new Corollary IX.

1713, the language of proportion is defined, and an error is corrected. A new Scholium after Lemma X states what “A is as B directly and C inversely” means, and Lemma XI defines the sesquiplicate ratio. Proposition IV’s Corollary 6 is corrected and generalized to any power of the radius.

1713, the Definitions and Laws. Definition V gains a long explanation (the sling, the projectile fired from a mountain, the Moon, and the inverse problems of mathematics). Corollary I of the Laws specifies impulses impressed at one place, and Law III is extended explicitly to attractions, with a new argument for the mutual gravity of the parts of the Earth.

1726, *products and quotients give way to compound ratios*. Where 1713 said “multiplied into” or “applied to”, 1726 often says “jointly” or “in the ratio compounded of” (Definition VIII’s explanation, Proposition IV’s Corollaries 1 and 2, the reflection argument after Proposition IV). The Scholium to the Laws gains an explanation of Galileo’s results by equal impulses in equal particles of time.

Definitions

Definition I. *The quantity of matter is its measure, arising from its density and its volume jointly.*

In modern terms: $m \propto \rho V$.

Air of double density in double the space is quadruple; likewise snow or powders condensed by compression or melting, and all bodies condensed by any cause. A medium freely pervading the interstices of the parts, if there is one, is disregarded. This quantity is meant below by *body* or *mass*. It is known through each body’s weight, to which it is proportional, as very accurate pendulum experiments show (shown later).

1713. The example adds: in triple the space, sextuple.

Definition II. *The quantity of motion is its measure, arising from the velocity and the quantity of matter jointly.*

In modern terms: $\mathbf{p} = m\mathbf{v}$.

The motion of the whole is the sum of the motions of the parts; a body twice as large with equal velocity has double the motion, and with double velocity quadruple.

Definition III. *The inherent force of matter (vis insita) is a power of resisting, by which every body, as far as it can, perseveres in its state of rest or of uniform motion in a straight line.*

It is always proportional to the body and differs from the inertia of mass only in the way it is conceived; hence it may be called the *force of inertia*. A body exerts it only when another force impressed on it changes its state. The exertion is, from different points of view, *resistance* (the body resists the impressed force to keep its state) and *impetus* (the body, yielding with difficulty to an obstacle, tries to change the obstacle’s state). Common usage gives resistance to bodies at rest and impetus to moving ones; but rest and motion, as commonly conceived, differ only relatively.

Definition IV. *Impressed force is an action exerted on a body to change its state of rest or of uniform motion in a straight line.*

It consists in the action alone and does not remain in the body afterwards; the body perseveres in any new state by the force of inertia alone. Impressed force has various origins: percussion, pressure, centripetal force.

Definition V. *Centripetal force is that by which a body is drawn, impelled, or in any way tends toward some point as to a centre.*

1713. *Bodies* (plural) are drawn, impelled or tend toward the point *from all sides*.

Examples: gravity, toward the centre of the Earth; magnetic force, by which iron seeks a magnet; and the force, whatever it is, by which the planets are continually drawn back from rectilinear motions and made to revolve in curved lines. Its quantity is of three kinds: absolute, accelerative and motive.

1713. The explanation is much expanded. A stone whirled in a sling endeavours to leave the hand, stretches the sling the more strongly the faster it revolves, and flies off when released; the contrary force by which the sling keeps drawing the stone back toward the hand, the centre of the orbit, is called centripetal. The same holds for all bodies moving in orbits: without a contrary force they would go off in straight lines with uniform motion. A projectile without gravity would go off in a straight line into the heavens; gravity bends it continually toward the Earth, more or less according to its gravity and velocity. A lead ball fired horizontally from a mountain top that lands two miles away would, with double or tenfold velocity, go about twice or ten times as far (air resistance removed); by increasing the velocity, the distance could be increased at will and the curvature of its path decreased, so that it would land ten, thirty or ninety degrees away, or go round the whole Earth before falling, or never fall and go off into the heavens. In the same way the Moon, by gravity or some other force urging it toward the Earth, can be drawn continually from its rectilinear course and bent into its orbit; with too small a force it would not be bent enough, with too great a force too much. It is for mathematicians to find the force that keeps a body exactly in a given orbit with a given velocity, and conversely to find the curved path into which a body leaving a given place with a given velocity is bent by a given force.

1726. “The less its gravity *for its quantity of matter*”, and the last alternative reads simply “or go off into the heavens”.

Definition VI. *The absolute quantity of a centripetal force is its measure, greater or less according to the efficacy of the cause propagating it from the centre through the surrounding regions.*

For instance, magnetic power is greater in one magnet and less in another.

1713. “according to the size of the magnet or the intensity of its power”.

Definition VII. *The accelerative quantity of a centripetal force is its measure proportional to the velocity it generates in a given time.*

In modern terms: at a given place, $a \propto \Delta v$ for a given Δt .

The same magnet acts more strongly at a smaller distance; gravity is greater in valleys and less on high mountain tops (pendulum experiments), and less still at greater distances from the Earth (shown later). At equal distances it is the same on all sides, because all falling bodies, heavy or light, large or small, are equally accelerated once air resistance is removed.

Definition VIII. *The motive quantity of a centripetal force is its measure proportional to the motion it generates in a given time.*

In modern terms: $F \propto \Delta p$ for a given Δt .

Weight is greater in a greater body and less in a smaller; in the same body, greater near the Earth and less in the heavens. This quantity is the centripetency of the whole body, its propensity toward the centre, its weight; it is always known by the equal and contrary force that can prevent the body's descent.

These quantities may be called briefly *absolute*, *accelerative* and *motive forces*, and referred respectively to the centre, to the place of the body, and to the body. The motive force belongs to the body as the endeavour of the whole toward the centre, compounded of the endeavours of all its parts; the accelerative force belongs to the place, as an efficacy diffused from the centre through each surrounding place to move the bodies there; the absolute force belongs to the centre, as endowed with some cause without which motive forces are not propagated, whether that cause is a central body or something else that does not appear. The concept is mathematical only; physical causes and seats of forces are not considered here.

Hence accelerative force is to motive force as velocity is to motion: the quantity of motion is velocity times quantity of matter, and motive force is accelerative force times the same quantity of matter.

In modern terms: $F = ma$ (as a proportionality).

1726. Motive force arises from the accelerative force *and* the quantity of matter *jointly*, and weight is as the body and the accelerative gravity *jointly*, where 1687 and 1713 say "multiplied into".

The sum of the actions of the accelerative force on the single particles of a body is the motive force of the whole. Near the Earth's surface, where accelerative gravity is the same in all bodies, weight is as the body; where accelerative gravity is smaller, weight decreases in proportion and is always as the body times accelerative gravity. Where accelerative gravity is half as large, a body half or a third as large weighs four or six times less.

Attractions and impulses are called accelerative and motive in the same sense. The words attraction, impulse or propensity toward a centre are used indifferently, the forces being considered mathematically and not physically. The reader should not take them to define a kind or mode of action or a physical cause, nor to attribute forces truly and physically to centres, which are mathematical points.

Scholium

Time, space, place and motion are not defined, being well known to all. But the common understanding conceives them only through their relation to sensible things, which breeds prejudices; to remove these, each is divided into absolute and relative, true and apparent, mathematical and common.

I. *Absolute, true and mathematical time*, of itself and by its nature without relation to anything external, flows equably, and is also called *duration*. *Relative, apparent and common time* is any sensible, external measure of duration by motion, accurate or not, used in place of true time: an hour, a day, a month, a year.

II. *Absolute space*, by its nature without relation to anything external, remains always similar and immovable. *Relative space* is any movable measure or dimension of

this space, determined by our senses through its position relative to bodies and commonly taken for immovable space, such as the dimension of a subterranean, aerial or celestial space fixed by its position relative to the Earth. Absolute and relative space are the same in kind and magnitude but do not always remain numerically the same: if the Earth moves, the space of our air, always the same relative to the Earth, is now one part of absolute space and now another.

- III. *Place* is the part of space a body occupies, absolute or relative according to the space. It is a part of space, not the position of the body or its surrounding surface: equal solids always have equal places, while their surfaces, for dissimilar figures, are mostly unequal; positions, properly speaking, have no quantity and are affections of places rather than places. Since the motion of the whole equals the sum of the motions of the parts, the place of the whole equals the sum of the places of the parts, and is internal and in the whole body.
- IV. *Absolute motion* is the translation of a body from one absolute place to another; *relative motion*, from one relative place to another. In a ship under sail, a body's relative place is the region of the ship it occupies, which moves with the ship; relative rest is remaining in that region; true rest is remaining in the same part of immovable space. If the Earth truly rests, a body resting relative to the ship moves truly with the ship's velocity on the Earth; if the Earth also moves, the true motion arises partly from the Earth's true motion in immovable space and partly from the ship's relative motion; and if the body moves within the ship, from all three. Example: if the Earth's surface there moves truly east with 10010 parts of velocity, the ship west with 10 parts, and a sailor walks east with 1 part, the sailor moves truly east with 10001 parts and relatively to the Earth west with 9 parts.

Absolute time is distinguished from relative time in astronomy by the equation of time: natural days are unequal, and astronomers correct this to measure celestial motions by a truer time. Perhaps no equable motion exists by which time can be measured exactly; all motions can speed up or slow down, but the flow of absolute time cannot change. Duration is the same whether motions are fast, slow or absent. The need for the equation is shown by pendulum clocks and by the eclipses of Jupiter's satellites.

As the order of the parts of time is immutable, so is the order of the parts of space. Times and spaces are, as it were, the places of themselves and of all things: all things are placed in time as to order of succession and in space as to order of position. It is of their essence to be places, and for primary places to move is absurd. These are absolute places, and only translations from them are absolute motions.

These parts of space cannot be seen or distinguished by our senses, so sensible measures are used instead: places are defined by positions and distances from some body regarded as immovable, and motions estimated with respect to those places. In human affairs this serves; in philosophy one must abstract from the senses, since possibly no body truly rests to which places and motions could be referred.

Absolute and relative rest and motion are distinguished by their properties, causes and effects.

Properties. Bodies truly at rest rest among themselves; but since some body far away, among or beyond the fixed stars, may rest absolutely while the positions of bodies in our regions cannot tell us whether any of them keeps a given position to it, true rest

cannot be defined from their mutual positions. Parts that keep given positions in their wholes share the wholes' motions: the parts of a revolving body endeavour to recede from the axis, and the impetus of a progressing body arises from the joint impetus of its parts. Hence when surrounding bodies move, bodies at rest relative to them move too, and true motion cannot be defined by translation from the neighbourhood of bodies regarded as resting; those bodies must truly rest. Likewise, when a place moves, what is in it moves with it; a body moving from a moving place shares its place's motion. Every whole and absolute motion is compounded of the motion of the body from its first place, of that place from its place, and so on, until an immovable place is reached. Whole and absolute motions can therefore be defined only by immovable places, and immovable places are only those that keep given positions to each other from infinity to infinity, which constitute immovable space.

Causes. True motion is neither generated nor changed except by forces impressed on the moving body itself; relative motion can be generated or changed without forces impressed on it, by forces impressed only on the bodies of reference. True motion always changes under forces impressed on the moving body; relative motion need not, if the same forces act on the bodies of reference so as to keep the relative position.

Effects. The forces of receding from the axis of circular motion distinguish absolute from relative motion: in purely relative circular motion they are null, in true motion greater or less according to the quantity of motion. A bucket hung by a long cord is turned until the cord is stiffly twisted, filled with water, and released. At first the surface of the water is flat, as before the motion; as the bucket gradually communicates its motion to the water, the water recedes from the middle and rises at the sides, becoming concave, and rises more as its motion grows, until it revolves in equal times with the bucket and rests relative to it. The rise shows the endeavour to recede from the axis, and by it the true and absolute circular motion of the water is known and measured, quite contrary to the relative motion: at the start, when the relative motion was greatest, there was no such endeavour and no true circular motion.

Relative quantities are therefore not the quantities whose names they bear but their sensible measures. Those who read these words in Scripture as the measured quantities do it violence, and those who confuse true quantities with their relations and common measures corrupt mathematics and philosophy.

To know the true motions of single bodies and distinguish them from apparent ones is very difficult, because the parts of immovable space do not come under the senses. But the matter is not hopeless: arguments are available partly from apparent motions, which are differences of true motions, and partly from forces, which are causes and effects of true motions. Two globes joined by a cord at a given distance and revolving about their common centre of gravity show, by the cord's tension, their endeavour to recede from the axis, from which the quantity of circular motion can be computed. Impressing equal forces on alternate faces to increase or decrease the motion, the change in tension shows which faces must be pushed to increase the motion most, namely the trailing faces, and hence the direction of the motion. In this way the quantity and direction of the circular motion could be found even in an immense vacuum with nothing external and sensible to compare. How true motions are inferred from their causes, effects and apparent differences, and conversely, is taught more fully in what follows; the treatise was composed for this purpose.

Axioms, or Laws of Motion

Law I. *Every body perseveres in its state of rest or of uniform motion in a straight line, except insofar as it is compelled by impressed forces to change that state.*

In modern terms: with no impressed force, $\mathbf{v}$ is constant.

Projectiles persevere in their motions except as air resistance retards them and gravity impels them downward. A top, whose cohering parts continually draw each other back from rectilinear motions, keeps spinning except as the air retards it. The greater bodies of planets and comets keep their progressive and circular motions longer in less resisting spaces.

Law II. *The change of motion is proportional to the motive force impressed, and takes place along the straight line in which that force is impressed.*

In modern terms: $\Delta \mathbf{p} \propto \mathbf{J}$, with $\mathbf{J}$ the impressed motive force (an impulse) and $\Delta \mathbf{p}$ parallel to it.

If a force generates some motion, double the force generates double the motion and triple the force triple, whether impressed all at once or gradually and successively. This motion, always directed with the generating force, is added to the body's earlier motion if they conspire, subtracted if contrary, and compounded obliquely if oblique, according to both directions.

Law III. *To every action there is always an equal and contrary reaction; or, the mutual actions of two bodies on each other are always equal and directed to contrary parts.*

In modern terms: $\mathbf{F}_{AB} = -\mathbf{F}_{BA}$.

Whatever presses or pulls another is pressed or pulled by it as much. If you press a stone with your finger, the finger is pressed by the stone. If a horse pulls a stone tied to a rope, the horse is pulled back equally toward the stone: the rope, stretched both ways, urges each toward the other with the same endeavour to relax. If a body striking another changes its motion, it undergoes the same change of its own motion in the contrary direction. The equal changes are of motions, not velocities (for bodies not otherwise impeded); the changes of velocities, also contrary, are inversely proportional to the bodies.

In modern terms: $m_A \Delta \mathbf{v}_A = -m_B \Delta \mathbf{v}_B$.

1713. The explanation ends: this law also holds in attractions, as will be proved in the next Scholium.

Corollary I. *A body acted on by two forces jointly describes the diagonal of a parallelogram in the same time in which it would describe the sides by the forces separately.*

Proof. Let the body in a given time be carried by force M alone from A to B , and by force N alone from A to C ; complete the parallelogram $ABDC$. Since N acts along AC , parallel to BD , it does not change the velocity of approach to the line BD generated by M . So the body reaches the line BD in the same time whether N acts or not, and at the end of that time it is somewhere on BD . By the same argument it is somewhere on CD , so it is at their intersection D .

In modern terms: $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC}$.

1713. The forces M and N are *impressed at the place A*, the motion from A to B is *uniform*, the invariance of the velocity of approach is by *Law II*, and the proof ends: the body proceeds in a straight line from A to D by *Law I*. The corollary is thereby stated for impulses given at one place, followed by inertial motion.

Corollary II. *Hence the composition of a direct force AD out of any oblique forces AB and BD , and conversely the resolution of any direct force AD into any oblique forces AB and BD . This is abundantly confirmed by mechanics.*

Weights A and P hung by threads from unequal spokes OM , ON of a wheel balance when they are inversely as the perpendicular distances OK , OL of their threads from the centre (the property of the balance, the lever and the wheel and axle): resolving the weight A , represented by AD , into AC along the spoke OD , which does nothing to turn the wheel, and DC perpendicular to it, gives $P : A = DC : DA = OK : OL$ by the similar triangles ADC , DOK . A weight p resting partly on an inclined plane and partly hung by a thread resolves likewise, which gives the tensions of oblique threads, the forces of the wedge between the faces of a split body, of the mallet, and of the screw, which is a wedge driven by a lever. From these follow the forces of all machines built from wheels, drums, pulleys, levers, taut strings and weights, and of the muscles moving the bones of animals.

Corollary III. *The quantity of motion, obtained by taking the sum of the motions directed to the same side and the difference of those directed to contrary sides, is not changed by the actions of bodies on each other.*

In modern terms: $\sum_i m_i \mathbf{v}_i$ is invariant under mutual interactions.

Proof. Action and reaction are equal (*Law III*), so by *Law II* they produce equal changes of motion toward contrary sides. If the motions are to the same side, whatever is added to the motion of the body fleeing is taken from the body pursuing, so the sum stays the same; if the bodies meet, equal amounts are taken from both, so the difference stays the same.

Example: a sphere A three times as large as a sphere B moves with 2 parts of velocity, and B follows on the same line with 10, so the motions are 6 and 10, total 16. Whatever A gains in the collision, 3, 4 or 5 parts, B loses, leaving A with 9, 10 or 11 and B with 7, 6 or 5; if A gains 9 to 12, B continues with 1, stops, or recoils with 1 or 2. The sum of conspiring motions or the difference of contrary ones is always 16. For non-spherical bodies or oblique collisions, find the plane touching both at the point of contact, resolve each motion (*Corollary II*) into components perpendicular and parallel to that plane, keep the parallel components, and give the perpendicular components equal and contrary changes so that the sum or difference is unchanged. Collisions also produce rotations about the bodies' own centres, which are not considered here.

Corollary IV. *The common centre of gravity of two or more bodies does not change its state of motion or rest by the actions of the bodies among themselves; therefore the common centre of gravity of all bodies acting on each other (external actions and impediments excluded) either rests or moves uniformly in a straight line.*

In modern terms: with no external forces, $\mathbf{R} = \sum_i m_i \mathbf{r}_i / \sum_i m_i$ satisfies $\ddot{\mathbf{R}} = 0$.

1713. "The common centre of gravity of two or more bodies". The proof

treats points moving in one plane and adds that the same holds if their motions are not in one plane. **1726** cites Lemma XXIII *and its corollary*.

Proof. If two points move uniformly in straight lines and their distance is divided in a given ratio, the dividing point rests or moves uniformly in a straight line (proved later in Lemma XXIII for the plane; the same holds in space). So if any number of bodies move uniformly in straight lines, the common centre of any two rests or moves uniformly, since it divides in a given ratio the line joining their centres; likewise the common centre of these two and any third, and so on to infinity. Hence in a system of bodies free of mutual actions and of all external forces, each moving uniformly in a straight line, the common centre of all rests or moves uniformly in a straight line. Further, for two bodies acting on each other, their distances from the common centre are inversely as the bodies, so their relative motions of approach to or recession from the centre are equal; the centre is neither advanced nor retarded by equal changes of motion in contrary directions, that is, by the mutual actions. In a system of many bodies, the common centre of any two acting on each other keeps its state, and the common centre of the rest is unaffected by that action; the distance between these two centres is divided by the common centre of all inversely as the total bodies whose centres they are; so the common centre of all keeps its state. All mutual actions are between pairs of bodies or compounded of such, so the common centre of all is never changed by them.

Corollary V. *The motions of bodies contained in a given space are the same among themselves whether that space rests or moves uniformly in a straight line without circular motion.*

Proof. The differences of motions toward the same side and the sums of motions toward contrary sides are the same at the start in both cases (by hypothesis), and from them arise the collisions and impulses by which the bodies strike each other; so by Law II the effects are equal in both cases, and the motions among themselves stay equal. A clear experiment confirms it: all motions behave the same in a ship at rest or moving uniformly in a straight line.

In modern terms: the laws are invariant under $\mathbf{r}' = \mathbf{r} - \mathbf{u}t$, $t' = t$, with $\mathbf{u}$ constant.

Corollary VI. *If bodies move in any way among themselves and are urged along parallel lines by equal accelerative forces, they all continue to move among themselves as if those forces did not act.*

Proof. The forces, acting equally (in proportion to the quantities of the bodies) along parallel lines, move all bodies equally as to velocity (Law II), and so never change their positions and motions among themselves.

Scholium

These principles are received by mathematicians and confirmed by many experiments. By the first two Laws and Corollaries Galileo found that the descent of heavy bodies is in the duplicate ratio of the time and that the motion of projectiles is in a parabola, as experiment agrees except for the small retardation by air. On the same Laws and Corollaries depend the demonstrations about the times of oscillating pendulums, confirmed daily by clocks. From these and the third Law, Wren, Wallis and Huygens independently

found the rules of the collision and reflection of two bodies and communicated them to the Royal Society at about the same time, in full agreement. Wren confirmed them by an experiment with pendulums before the Society, and Mariotte soon devoted a whole book to it. For exact agreement one must allow for the air's resistance and for the elastic force of the colliding bodies. Newton describes how to correct a pendulum-collision experiment for air resistance by measuring the loss over swings, and reports his own trials with bodies of various hardness, including tightly wound wool balls: the bodies separate with a relative velocity in a given ratio to the relative velocity of approach, and the rules hold for soft bodies as well as hard ones.

1713. The description of the pendulum experiment is made exact: the arcs are marked so that $RS = TV$ and $RS : ST = 3 : 2$, and the equal changes of motion are stated for the bodies.

1726. A passage is inserted after the mention of Galileo. Uniform gravity, acting equally in equal particles of time, impresses equal forces on a falling body and generates equal velocities; in the whole time it impresses the whole force and generates the whole velocity, proportional to the time; and the spaces described in proportional times are as the velocities and times jointly, that is, in the duplicate ratio of the times. For a body thrown upward, uniform gravity impresses forces and takes away velocities proportional to the times; the times of ascent to the greatest heights are as the velocities to be taken away, and the heights as the velocities and times jointly, or in the duplicate ratio of the velocities. A body thrown along any straight line compounds the motion from the projection with the motion from gravity: if by projection alone it would describe AB in a given time and by falling alone the height AC , completing the parallelogram $ABDC$ puts it at D at the end of the time, and the curve AED it describes is a parabola touching AB at A whose ordinate BD is as AB^2 .

For attractions, suppose two bodies A and B attract each other and an obstacle is placed between them. If A were drawn toward B more than B toward A , the obstacle would be pressed more by A than by B and would not stay in equilibrium; the stronger pressure would make the system of the two bodies and the obstacle move toward B and, accelerating forever in free space, go off to infinity, which is absurd and contrary to the first Law. So the bodies press the obstacle equally and attract each other equally. Newton tried this with a lodestone and iron floating in separate vessels on still water: neither pushed the other, and they came to rest in equilibrium.

1713. A new paragraph: gravity between the Earth and its parts is mutual. Cut the Earth by any plane EG into parts EGF and EGI . Cut the larger part EGI by a parallel plane HK into $EGKH$ and HKI , with HKI equal to EGF . The middle part $EGKH$ inclines by its own weight toward neither extreme part and rests between them in equilibrium; the extreme part HKI presses with its whole weight on the middle part and urges it toward EGF . So the force with which EGI tends toward EGF equals the weight of HKI , that is, of EGF , and the two parts weigh equally on each other. Otherwise the whole Earth, floating in the free aether, would yield to the greater weight and go off to infinity.

As bodies are equipollent in collision when their velocities are inversely as their inher-

ent forces, so in mechanical instruments agents are equipollent and sustain each other when their velocities, estimated along the forces, are inversely as the forces. So for the balance, the pulley, trains of wheels, the screw and the wedge. The use of machines lies only in increasing force by diminishing velocity and conversely; this solves, for every suitable instrument, the problem of moving a given weight by a given force. If the action of the agent is estimated by its force and velocity jointly, and the reaction of the resistance by the velocities of its parts and their forces of resisting (from friction, cohesion, weight and acceleration), action and reaction are always equal in every use of instruments. Mechanics is not the subject here; this only shows how widely and how surely the third Law holds.

Book I: On the Motion of Bodies

Section I. On the method of first and last ratios, by whose help what follows is demonstrated

Lemma I. *Quantities, and also ratios of quantities, which in a given finite time tend constantly to equality, and before the end of that time approach nearer to each other than by any given difference, become ultimately equal.*

In modern terms: if $X(t), Y(t)$ on $[0, T)$ approach each other so that for every $\varepsilon > 0$ there is $t < T$ beyond which $|X - Y| < \varepsilon$, then their ultimate values at T are equal.

Proof. If not, let their ultimate difference be $D > 0$. Then they cannot approach nearer to equality than by the given difference D , contrary to the hypothesis.

1713. The statement reads “in any finite time” and replaces “can approach nearer” by “before the end of that time approach nearer than by any given difference”: the approach is asserted to happen within the time, not merely to be possible. The proof begins “If you deny it, let them be ultimately unequal”.

Lemma II. *If in any figure $AacE$, bounded by the straight lines Aa , AE and the curve acE , any number of parallelograms Ab , Bc , Cd , ... are inscribed on equal bases AB , BC , CD , ... with sides Bb , Cc , Dd , ... parallel to the side Aa , and the parallelograms $aKbl$, $bLcm$, $cMdn$, ... are completed; and if then the width of the parallelograms is diminished and their number increased indefinitely: the ultimate ratios which the inscribed figure $AKbLcMdD$, the circumscribed figure $AalbmcndoE$ and the curvilinear figure $AabcdE$ have to one another are ratios of equality.*

In modern terms: for the region under a monotone curve over $[a, b]$, with lower and upper step sums L_n , U_n on n equal subintervals and area $\mathcal{A}$, the ratios $L_n : U_n : \mathcal{A}$ tend to $1 : 1 : 1$. [In Newton’s figure the curve is monotone, and the curvilinear figure’s area is taken as given.]

Proof. The difference of the inscribed and circumscribed figures is the sum of the parallelograms $Kl + Lm + Mn + Do$, that is (the bases being equal) the rectangle under one base Kb and the sum of the heights Aa , the rectangle $ABla$. Since its width AB is diminished indefinitely, it becomes less than any given rectangle. So by Lemma I the inscribed and circumscribed figures, and still more the curvilinear figure between them, become ultimately equal.

In modern terms: $U_n - L_n = \Delta x (f(a) - f(b)) \rightarrow 0$.

Lemma III. *The same ultimate ratios are also ratios of equality when the widths AB , BC , CD , ... of the parallelograms are unequal and all are diminished indefinitely.*

Proof. Let AF equal the greatest width and complete the parallelogram $FAaf$. It is greater than the difference of the inscribed and circumscribed figures, and as its width AF is diminished indefinitely it becomes less than any given rectangle.

In modern terms: $U - L \leq \max_i \Delta x_i (f(a) - f(b)) \rightarrow 0$ as the mesh tends to 0.

Corollary 1. Hence the ultimate sum of the vanishing parallelograms coincides in every part with the curvilinear figure.

Corollary 2. Still more does the rectilinear figure bounded by the chords ab , bc , cd , ... of the vanishing arcs coincide ultimately with the curvilinear figure.

Corollary 3. So does the rectilinear figure circumscribed by the tangents of the same arcs.

Corollary 4. And therefore these ultimate figures (as to their perimeters acE) are not rectilinear, but curvilinear limits of rectilinear figures.

Lemma IV. *If in two figures $AacE$, $PprT$ two series of parallelograms are inscribed (as above), equal in number, and when the widths are diminished indefinitely the ultimate ratios of the parallelograms of one figure to those of the other, each to each, are the same: the two figures are to each other in that same ratio.*

Proof. As the parallelograms are to each other one by one, so (*componendo*) is the sum of all to the sum of all, and so is figure to figure, each figure being to its sum in the ratio of equality (Lemma III).

In modern terms: if $a_i/b_i \rightarrow r$ for all i as the mesh tends to 0, then $\lim \sum a_i / \lim \sum b_i = r$. [The convergence is tacitly uniform in i .]

Corollary. Hence if two quantities of any kind are divided into the same number of parts, and those parts, as their number is increased and their magnitude diminished indefinitely, have a given ratio to each other, first to first, second to second and so on in order, the wholes are to each other in that given ratio. For if parallelograms are taken in the figures of this Lemma proportional to the parts, the sums of the parts are always as the sums of the parallelograms, and so, in the limit, in the ultimate ratio of parallelogram to parallelogram, that is (by hypothesis) of part to part.

Lemma V. *All corresponding sides of similar figures, curvilinear as well as rectilinear, are proportional, and the areas are in the duplicate ratio of the sides.*

In modern terms: under a similarity of ratio λ , lengths scale by λ and areas by λ^2 .

Lemma VI. *If any arc AB , given in position, is subtended by its chord AB , and at some point A in the middle of a continuous curvature it is touched by a straight line AD produced both ways; and if then the points A and B approach each other and coincide: the angle BAD contained by the chord and the tangent is diminished indefinitely and ultimately vanishes.*

In modern terms: at a point where the curve has a tangent, the direction of the chord AB tends to the direction of the tangent as $B \rightarrow A$.

Proof. Produce AB to b and AD to d . As A and B coincide, no part AB of the line Ab lies any longer inside the curve, so the line Ab either coincides with the tangent Ad or passes between the tangent and the curve. The latter is contrary to the nature of curvature, so the former holds.

1713. A new proof replaces this one: if the angle does not vanish, the arc AB will contain with the tangent AD an angle equal to a rectilinear angle, and so the curvature at A will not be continuous, contrary to the hypothesis. **1726** names the arc ACB , through an intermediate point C , here and in Lemmas VII and VIII.

Lemma VII. *Under the same suppositions, the ultimate ratio of the arc, the chord and the tangent to each other is the ratio of equality.*

Proof. Produce AB and AD to b and d , draw bd parallel to the secant BD , and let the arc Ab be similar to the arc AB . As A and B coincide, the angle dAb vanishes (Lemma VI), so the straight lines Ab , Ad and the arc between them coincide and are equal. Hence AB , AD and the arc AB , always proportional to these, ultimately have the ratio of equality.

In modern terms: $\text{arc } AB / \text{chord } AB \rightarrow 1$ and $AD / \text{chord } AB \rightarrow 1$ as $B \rightarrow A$.

1713. The proof is reorganized as a rescaling: while B approaches A , the lines AB , AD are imagined always produced to *distant* points b , d , so that the enlarged lines Ab , Ad and the arc between them stay *always finite* while the angle dAb vanishes; they coincide and are equal, and the proportional small lines AB , AD and the arc AB *vanish* with ultimate ratio of equality.

Corollary 1. If through B a line BF is drawn parallel to the tangent, meeting any straight line AF through A in F , then BF ultimately has the ratio of equality to the vanishing arc AB , because, completing the parallelogram $AFBD$, it always has that ratio to AD .

Corollary 2. If through B and A more lines BE , BD , AF , AG are drawn cutting the tangent AD and its parallel BF , the ultimate ratio of all the abscissas AD , AE , BF , BG , and of the chord and arc AB , to each other is the ratio of equality.

Corollary 3. Therefore all these lines may be used for one another in any argument about ultimate ratios.

Lemma VIII. *If the given straight lines AR , BR form with the arc AB , the chord AB and the tangent AD three triangles RAB , RAB , RAD (the first with the arc as a side), and then the points A and B approach each other: the ultimate form of the vanishing triangles is that of similarity, and their ultimate ratio is that of equality.*

Proof. Produce AB , AD , AR to b , d , r ; draw rbd parallel to RD , and the arc Ab similar to the arc AB . As A and B coincide, the angle bAd vanishes, so the three triangles rAb , rAb , rAd coincide and are therefore similar and equal. Hence RAB , RAB , RAD , always similar and proportional to these, become ultimately similar and equal to each other.

1713. The same rescaling: AB , AD , AR are always produced to distant points b , d , r , and the three triangles rAb , rAb , rAd stay *always finite* while they coincide.

Corollary. Hence those triangles may be used for one another in any argument about ultimate ratios.

Lemma IX. *If a straight line AE and a curve AC , given in position, cut each other at a given angle A , and to that line, at another given angle, ordinates BD , EC are applied meeting the curve in B , C , and then the points B , C approach the point A : the areas of the triangles ADB , AEC will be ultimately to each other in the duplicate ratio of the sides.*

In modern terms: $\text{area}(ADB)/\text{area}(AEC) \rightarrow (AD/AE)^2$ as $B, C \rightarrow A$, the triangles having the curved side.

Proof. On AD produced take Ad , Ae proportional to AD , AE , and erect ordinates db , ec parallel and proportional to DB , EC . Produce AC to c , draw the curve Abc similar to ABC , and the straight line Ag touching both curves at A and cutting the ordinates in F , G , f , g . Then let B and C coincide with A : the angle cAg vanishes, the curvilinear areas Abd , Ace coincide with the rectilinear Afd , Age , and so (Lemma V) are in the duplicate ratio of the sides Ad , Ae . But the areas ABD , ACE are always proportional to these, and the sides AD , AE to these sides. So the areas ABD , ACE are ultimately in the duplicate ratio of the sides AD , AE .

1713. The enlarged figure is again kept finite: while B , C approach A , the line AD is always produced to distant points d , e , and the points B , C coalesce with A while the length Ae stays fixed.

Lemma X. *The spaces which a body describes, urged by any regular force, are at the very beginning of the motion in the duplicate ratio of the times.*

1713. *The spaces which a body describes, urged by any finite force, whether that force is determinate and immutable or is continually increased or continually decreased, are at the very beginning of the motion in the duplicate ratio of the times.* The 1687 “regular force” becomes a finite force that is constant or monotone; Corollaries 1 and 2 are reworded accordingly (“by any equal forces”, “measured by the distances of the bodies from those places of the similar figures”).

In modern terms: $s(t_1)/s(t_2) \rightarrow (t_1/t_2)^2$ as $t_1, t_2 \rightarrow 0$; that is, $s(t) \sim c t^2$ at the start.

Proof. Represent the times by the lines AD , AE and the velocities generated by the ordinates DB , EC . The spaces described with these velocities are as the areas ABD , ACE described by these ordinates, that is, at the very beginning of the motion (Lemma IX), in the duplicate ratio of the times AD , AE .

In modern terms: $s(t) = \int_0^t v(\tau) d\tau$, the area under the velocity graph, whose initial piece is the curvilinear triangle of Lemma IX.

Corollary 1. Hence the errors of bodies describing similar parts of similar figures in proportional times, generated by equal forces similarly applied to the bodies in those parts, and measured from the places of the figures which the bodies would reach in the same proportional times without those forces, are very nearly as the squares of the times in which they are generated.

Corollary 2. The errors generated by proportional forces similarly applied are as the

forces and the squares of the times jointly.

In modern terms: the deviation from the force-free path is $\delta \approx \frac{1}{2}at^2$, so $\delta \propto Ft^2$ for a given body.

1713. Three new corollaries and a Scholium.

Corollary 3. The same holds for any spaces that bodies describe under different forces: at the very beginning of the motion they are as the forces and the squares of the times jointly, $s \propto Ft^2$.

Corollary 4. So the forces are as the spaces described at the very beginning directly and the squares of the times inversely, $F \propto s/t^2$.

Corollary 5. And the squares of the times are as the spaces directly and the forces inversely, $t^2 \propto s/F$.

Scholium. When indeterminate quantities of different kinds are compared, and one is said to be as another directly or inversely, the meaning is that the first increases or decreases in the same ratio as the second or as its reciprocal. When one is said to be as two or more others directly or inversely, it increases or decreases in the ratio compounded of the ratios in which the others, or their reciprocals, increase or decrease. So “ A is as B directly and C directly and D inversely” means that A changes in the same ratio as $B \cdot C/D$, that is, A and BC/D are in a given ratio.

Lemma XI. *The vanishing subtense of the angle of contact is ultimately in the duplicate ratio of the subtense of the conterminous arc.*

In modern terms: with BD the distance from B on the curve to the tangent at A , measured in a fixed direction, $BD \propto AB^2$ ultimately; for finite curvature κ , $BD \approx \frac{1}{2}\kappa AB^2$.

1713. The statement is restricted: *in all curves having finite curvature at the point of contact.*

Case 1. Let the arc be AB , its tangent AD , the subtense BD of the angle of contact perpendicular to the tangent, and AB the subtense of the arc. Erect AG perpendicular to the subtense AB and BG perpendicular to the tangent AD , meeting in G . Then let the points D, B, G go to d, b, g , and let I be the intersection of the lines BG, AG ultimately formed when D, B reach A . The distance GI can be made less than any assigned length. By the nature of circles through A, B, G and through A, b, g , $AB^2 = AG \cdot BD$ and $Ab^2 = Ag \cdot bd$, so the ratio $AB^2 : Ab^2$ is compounded of $AG : Ag$ and $BD : bd$. Since GI can be taken less than any assigned length, the ratio $AG : Ag$ can differ from equality by less than any assigned difference, and so $AB^2 : Ab^2$ can differ from $BD : bd$ by less than any assigned difference. By Lemma I the ultimate ratio $AB^2 : Ab^2$ equals the ultimate ratio $BD : bd$.

Case 2. Let BD be inclined to AD at any given angle; the ultimate ratio $BD : bd$ is the same as before, and so the same as $AB^2 : Ab^2$.

Case 3. Even if the angle D is not given, the angles D, d always tend to equality and approach each other nearer than by any assigned difference, so they are ultimately equal (Lemma I), and the lines BD, bd are in the same ratio as before.

1713. Case 3 also covers the line BD converging to a given point, or placed by any other law, provided the angles D, d are placed by a common law.

Corollary 1. Since the tangents AD, Ad , the arcs AB, Ab and their sines BC, bc become ultimately equal to the chords AB, Ab , their squares too are ultimately as the subtenses BD, bd .

1713. Two new corollaries follow, and the old Corollaries 2 and 3 become Corollaries 4 and 5.

Corollary 2. Their squares are also ultimately as the sagittas of the arcs that bisect the chords and converge to a given point, for those sagittas are as the subtenses BD, bd .

Corollary 3. Hence the sagitta is in the duplicate ratio of the time in which a body describes the arc with a given velocity.

Corollary 2. The rectilinear triangles ADB, Adb are ultimately in the triplicate ratio of the sides AD, Ad , and in the sesquiplicate ratio of the sides DB, db , being in the ratio compounded of AD and DB, Ad and db . Likewise the triangles ABC, Abc are ultimately in the triplicate ratio of the sides BC, bc .

1713 (now Corollary 4) adds a definition: the sesquiplicate ratio is the subduplicate of the triplicate, compounded of the simple and the subduplicate ratio, which others call sesquialteral; that is, the ratio of $3/2$ powers.

Corollary 3. Since DB, db are ultimately parallel and in the duplicate ratio of AD, Ad , the ultimate curvilinear areas ADB, Adb are (by the nature of the parabola) two thirds of the rectilinear triangles ADB, Adb , and the segments AB, Ab one third of the same triangles. Hence these areas and segments are in the triplicate ratio both of the tangents AD, Ad and of the chords and arcs AB, Ab .

Scholium

In all this we suppose the angle of contact neither infinitely greater than the angles of contact which circles make with their tangents, nor infinitely less; that is, the curvature at the point A neither infinitely small nor infinitely great, or the interval AI of finite magnitude. For DB can be taken as AD^3 , in which case no circle can be drawn through A between the tangent AD and the curve AB , and the angle of contact is infinitely less than circular ones. By a similar argument, taking DB successively as $AD^4, AD^5, AD^6, AD^7, \dots$ gives a series of angles of contact going on to infinity, each infinitely less than the one before. Taking DB successively as $AD^2, AD^{3/2}, AD^{4/3}, AD^{5/4}, AD^{6/5}, AD^{7/6}, \dots$ gives another infinite series, the first of the same kind as the circular ones, the second infinitely greater, and each infinitely greater than the one before. Between any two of these angles a series going on to infinity both ways can be inserted, each infinitely greater than the one before: for instance, between the terms AD^2 and AD^3 the series $AD^{13/6}, AD^{11/5}, AD^{9/4}, AD^{7/3}, AD^{5/2}, AD^{8/3}, AD^{11/4}, AD^{14/5}, AD^{17/6}, \dots$ And again between any two angles of this series a new series of intermediate angles can be inserted, differing from each other by infinite intervals. Nature knows no limit.

In modern terms: $DB \propto AD^p$ near A has curvature 0 for $p > 2$, finite nonzero curvature for $p = 2$, and infinite curvature for $1 < p < 2$; the orders p are dense.

What has been shown about curved lines and the surfaces they enclose applies easily to curved surfaces and contents of solids. These lemmas were set down first to avoid the tedium of long proofs by reduction to absurdity in the manner of the ancient geometers. Proofs are shorter by the method of indivisibles; but the hypothesis of indivisibles is harsher, and that method is therefore reckoned less geometrical. So the proofs of what follows are reduced to the ultimate sums and ratios of vanishing quantities and the first ratios of nascent ones, that is, to the limits of sums and ratios, and the proofs of those limits are given first as briefly as possible. This achieves what the method of indivisibles achieves, with principles now proved. Hence in what follows, whenever quantities are considered as made of particles, or small curved lines are used for straight ones, what is meant is not indivisibles but vanishing divisible quantities, not sums and ratios of determinate parts but always limits of sums and ratios, and the force of such proofs always rests on the method of the preceding lemmas.

Objection: vanishing quantities have no ultimate proportion, since before they vanish it is not ultimate, and when they have vanished there is none. By the same argument a body arriving at a place would have no ultimate velocity: before it arrives the velocity is not ultimate, and when it has arrived there is none. The answer is easy. The ultimate velocity is that with which the body moves neither before it reaches the last place and the motion stops, nor after, but at the moment it arrives. Likewise the ultimate ratio of vanishing quantities is the ratio with which they vanish, neither before nor after; the first ratio of nascent quantities is the ratio with which they begin; and the first and last sum is that with which they begin and cease to be (or to increase and decrease). There is a limit which the velocity can reach at the end of the motion but not exceed; this is the ultimate velocity. The same holds for the limit of all beginning and ceasing quantities and proportions. Since this limit is certain and definite, finding it is a truly geometrical problem, and geometrical things may legitimately be used to determine and prove other geometrical things.

Objection: if the ultimate ratios of vanishing quantities are given, their ultimate magnitudes are given too, and so every quantity consists of indivisibles, contrary to what Euclid proved about incommensurables in Book X of the *Elements*. This objection rests on a false hypothesis. The ultimate ratios with which quantities vanish are not ratios of ultimate quantities, but limits which the ratios of quantities decreasing without limit always approach, nearer than by any given difference, never passing them and never reaching them before the quantities are diminished indefinitely. This is clearer with infinitely great quantities: if two quantities with a given difference are increased indefinitely, their ultimate ratio, equality, is given, though no ultimate or greatest quantities of which it is the ratio are given. So in what follows, whenever for ease of imagination quantities are called least, vanishing or ultimate, do not understand quantities of determinate magnitude, but think of them as always to be diminished without limit.

In modern terms: an ultimate ratio is $\lim X/Y$, and Lemma I supplies its defining property.

Section II. On finding centripetal forces

Proposition I. Theorem I. *The areas which bodies made to move in orbits describe by radii drawn to an immovable centre of forces lie in immovable planes and are proportional to the times.*

In modern terms: for a centripetal force toward a fixed point S , the motion stays in one plane and $\frac{1}{2} \mathbf{r} \times \dot{\mathbf{r}}$ is constant, so swept area is proportional to time.

Proof. Divide the time into equal parts, and in the first part let the body describe the straight line AB by its inherent force. In the second part, if nothing impeded it, it would go straight on to c (Law I), describing Bc equal to AB , so that with radii AS , BS , cS drawn to the centre, the areas ASB and BSc would be equal. But when the body reaches B , let a centripetal force act with a single but great impulse and make the body deviate from the line Bc and go on in the line BC . Draw cC parallel to BS , meeting BC in C ; at the end of the second part of time the body (Corollary I of the Laws) will be at C , in the same plane as the triangle ASB . Join SC ; the triangle SBC equals the triangle SBC , because of the parallels SB , Cc , and so it also equals the triangle SAB . By a similar argument, if the centripetal force acts successively at C , D , E , ..., making the body describe in the single particles of time the single straight lines CD , DE , EF , ..., these lie in the same plane, and the triangle SCD equals the triangle SBC , SDE equals SCD , and SEF equals SDE . So in equal times equal areas are described in an immovable plane, and *componendo*, any sums of areas $SADS$, $SAFS$ are to each other as the times of description. Now let the number of triangles be increased and their width diminished indefinitely: their ultimate perimeter ADF (Corollary 4 of Lemma III) will be a curved line; so the centripetal force, by which the body is perpetually drawn back from the tangent of this curve, will act uninterruptedly, and any areas described, $SADS$, $SAFS$, being always proportional to the times of description, will be proportional to those times in this case too.

In modern terms: with vertices $\mathbf{r}_k$ and impulses toward S , $\mathbf{r}_{k+1} = 2\mathbf{r}_k - \mathbf{r}_{k-1} + \lambda_k \mathbf{r}_k$. Then $\mathbf{r}_k \times \mathbf{r}_{k+1} = \mathbf{r}_{k-1} \times \mathbf{r}_k$, so every triangle $S\mathbf{r}_k\mathbf{r}_{k+1}$ has the same area (Newton: triangles on the same base between parallels, Euclid I.37–38). The final step passes to the limit of the polygons.

Corollary 1. In non-resisting media, if the areas are not proportional to the times, the forces do not tend to the point where the radii meet.

Corollary 2. In all media, if the description of areas is accelerated, the forces do not tend to the point where the radii meet, but decline from it *in consequentia*.

1713. These two corollaries move, expanded, to Proposition II. In their place come six new ones.

Corollary 1. The velocity of a body attracted to an immovable centre is, in non-resisting spaces, inversely as the perpendicular let fall from the centre onto the rectilinear tangent of the orbit; for the velocities at A , B , C , D , E are as the bases AB , BC , CD , DE , EF of equal triangles, and these bases are inversely as the perpendiculars on them. In modern terms: $v \cdot p$ is constant, with p the distance from S to the tangent.

Corollary 2. If the chords AB , BC of two arcs described successively in equal times by the same body in non-resisting spaces are completed into a parallelogram $ABCU$, its diagonal BU , in the position it has ultimately when the arcs are diminished indefinitely, passes through the centre of forces when produced both ways.

Corollary 3. If the chords AB , BC and DE , EF of arcs described in equal

times are completed into parallelograms $ABCU$, $DEFZ$, the forces at B and E are in the ultimate ratio of the diagonals BU , EZ as the arcs are diminished indefinitely; for the motions BC , EF are compounded (Corollary I of the Laws) of Bc , BU and Ef , EZ , and BU , EZ , equal to Cc , Ff , were generated in the proof of this Proposition by the impulses of the centripetal force at B and E , and so are proportional to those impulses.

Corollary 4. The forces by which any bodies in non-resisting spaces are drawn from rectilinear motions and bent into curved orbits are to each other as the sagittas of arcs described in equal times that converge to the centre of forces and bisect the chords, as the arcs are diminished indefinitely; for these sagittas are halves of the diagonals of Corollary 3.

Corollary 5. So these forces are to the force of gravity as these sagittas are to the vertical sagittas of the parabolic arcs that projectiles describe in the same time.

Corollary 6. All the same holds, by Corollary V of the Laws, when the planes in which the bodies move, together with the centres of forces in them, are not at rest but move uniformly in a straight line.

Proposition II. Theorem II. *Every body that moves in some curved line and, by a radius drawn to a point either immovable or moving uniformly in a straight line, describes about that point areas proportional to the times, is urged by a centripetal force tending to that same point.*

In modern terms: if $\mathbf{r} \times \dot{\mathbf{r}}$ is constant about S , the force is parallel to $\mathbf{r}$.

Case 1. Every body moving in a curved line is turned from a rectilinear course by some force acting on it (Law I). The force by which the body is turned from its rectilinear course and made to describe, about the immovable point S , the least triangles SAB , SBC , SCD , ... equal in equal times, acts at the place B along a line parallel to cC (Euclid I.40 and Law II), that is, along the line BS ; and at the place C along a line parallel to dD , that is, along the line CS ; and so on. It therefore always acts along lines tending to that immovable point S .

Case 2. By Corollary V of the Laws it makes no difference whether the surface on which the body describes the curvilinear figure rests, or moves uniformly in a straight line together with the body, the figure described and its point S .

1713. The statement specifies a curved line *described in a plane*. Two corollaries are added, rewritten from the old corollaries of Proposition I.

Corollary 1. In non-resisting spaces or media, if the areas are not proportional to the times, the forces do not tend to the point where the radii meet, but decline from it *in consequentia*, toward the direction of motion, if the description of areas is accelerated, and *in antecedentia* if it is retarded.

Corollary 2. In resisting media too, if the description of areas is accelerated, the directions of the forces decline from the point where the radii meet toward the direction of motion.

Scholium

A body may be urged by a centripetal force compounded of several forces. In that case the meaning of the Proposition is that the force compounded of them all tends to the point S . Further, if some force acts along a line perpendicular to the surface described, it will make the body deviate from the plane of its motion, but it will neither increase nor diminish the quantity of the surface described, and so it may be neglected in the composition of forces.

Proposition III. Theorem III. *Every body that, by a radius drawn to the centre of another body moving in any way, describes about that centre areas proportional to the times, is urged by a force compounded of the centripetal force tending to that other body and of all the accelerative force by which that other body is urged.*

In modern terms: if $\mathbf{r}_{12} \times \dot{\mathbf{r}}_{12}$ is constant for the relative position $\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2$, then $\mathbf{a}_1 - \mathbf{a}_2$ is parallel to $\mathbf{r}_{12}$.

Proof. By Corollary VI of the Laws, if both bodies are urged along parallel lines by a new force equal and contrary to the force urging the other body, the first body continues to describe the same areas about the other as before. The force urging the other body is now destroyed by its equal and contrary force, so (Law I) that other body either rests or moves uniformly in a straight line, and the first body, urged by the difference of the forces, continues to describe areas proportional to the times about it. So (Theorem II) the difference of the forces tends to that other body as its centre.

1713. The proof names the bodies L and T ; T , left to itself once its force is cancelled, rests or moves uniformly, and L moves under “the difference of the forces, that is, the remaining force”. The argument is unchanged.

Corollary 1. Hence if one body, by a radius drawn to another, describes areas proportional to the times, and from the total force urging the first (simple, or compounded of several forces by Corollary II of the Laws) the whole accelerative force urging the other is subtracted (by the same Corollary), all the remaining force urging the first tends to the other body as centre.

Corollary 2. And if those areas are very nearly proportional to the times, the remaining force tends very nearly to the other body.

Corollary 3. And conversely, if the remaining force tends very nearly to the other body, those areas are very nearly proportional to the times.

Corollary 4. If a body, by a radius drawn to another body, describes areas that are very unequal compared with the times, and that other body rests or moves uniformly in a straight line, the action of the centripetal force tending to that other body is either null or mixed and compounded with very powerful actions of other forces; and the total force compounded of them all, if there are several, is directed to another centre (immovable or moving) about which the description of areas is uniform. The same holds when the other body moves in any way, provided the centripetal force taken is what remains after subtracting the total force acting on that other body.

Scholium

Since the uniform description of areas indicates the centre to which the force that most affects the body is directed, and the body is retained in its orbit by a force toward that centre, and every circular motion is rightly said to be performed about the centre by whose force the body is drawn back from rectilinear motion and retained in its orbit: why should we not use the uniform description of areas, in what follows, as the index of the centre about which every circular motion in free spaces is performed?

Proposition IV. Theorem IV. *The centripetal forces of bodies describing different circles with uniform motion tend to the centres of those circles, and are to each other as the squares of the arcs described in the same time applied to the radii of the circles.*

In modern terms: for arcs s described in a common time on circles of radius r , $F \propto s^2/r$; that is, $F \propto v^2/r$.

Proof. Let the bodies B, b , revolving in the circumferences of the circles BD, bd , describe the arcs BD, bd in the same time. By their inherent force alone they would describe tangents BC, bc equal to these arcs; so it is the centripetal forces that perpetually draw the bodies back from the tangents to the circumferences, and they are to each other in the first ratio of the nascent spaces CD, cd . They tend to the centres of the circles by Theorem II, because the areas described by the radii are taken proportional to the times. Make the figure tkb similar to the figure DCB . By Lemma V the line CD is to the line kt as the arc BD to the arc bt ; by Lemma XI the nascent line tk is to the nascent line dc as bt^2 to bd^2 ; so *ex aequo* the nascent line DC is to the nascent line dc as $BD \cdot bt$ to bd^2 , or equivalently as $BD \cdot bt/Sb$ to bd^2/Sb , and so (the ratios bt/Sb and BD/SB being equal) as BD^2/SB to bd^2/Sb .

In modern terms: the departure from the tangent over the arc s is the sagitta $CD \approx s^2/(2r)$ (Lemma XI), and the force is as that departure in a given time (Lemma X, Corollary 2).

1713. A new proof replaces this one. The forces tend to the centres of the circles by Proposition II and Corollary 2 of Proposition I, and are to each other as the versed sines of the least arcs described in equal times (Corollary 4 of Proposition I), that is, as the squares of those arcs applied to the diameters of the circles (Lemma VII). Since these arcs are as the arcs described in any equal times, and the diameters as the radii, the forces are as the squares of arcs described in the same time applied to the radii.

Corollary 1. Hence the centripetal forces are as the squares of the velocities applied to the radii of the circles: $F \propto v^2/r$.

Corollary 2. And they are inversely as the squares of the periodic times applied to the radii: $F \propto r/T^2$. That is, in the language of the geometers, these forces are in the ratio compounded of the duplicate ratio of the velocities directly and the simple ratio of the radii inversely; and also in the ratio compounded of the simple ratio of the radii directly and the duplicate ratio of the periodic times inversely.

Corollary 3. So if the periodic times are equal, both the centripetal forces and the velocities are as the radii, and conversely.

Corollary 4. If the squares of the periodic times are as the radii, the centripetal forces

are equal and the velocities are in the subduplicate ratio of the radii, and conversely: $T^2 \propto r \iff F = \text{const}, v \propto r^{1/2}$.

Corollary 5. If the squares of the periodic times are as the squares of the radii, the centripetal forces are inversely as the radii and the velocities equal, and conversely: $T \propto r \iff F \propto 1/r, v = \text{const}$.

Corollary 6. If the squares of the periodic times are as the cubes of the radii, the centripetal forces are inversely as the squares of the radii, and the velocities in the subduplicate ratio of the radii; and conversely: $T^2 \propto r^3 \iff F \propto 1/r^2$. [As printed in 1687 the velocity clause lacks “inversely”; by Corollary 1 the consistent relation is $v \propto r^{-1/2}$.]

Corollary 7. All the same holds for the times, velocities and forces with which bodies describe similar parts of any similar figures having similarly placed centres, by applying the preceding proof to those cases.

1713. The corollaries are rewritten and extended to nine.

Corollary 1. Therefore, since those arcs are as the velocities of the bodies, the forces are as the squares of the velocities applied to the radii, or, as the geometers say, in the ratio compounded of the duplicate ratio of the velocities directly and the simple ratio of the radii inversely.

Corollary 2. And since the periodic times are in the ratio compounded of the radii directly and the velocities inversely, the forces are inversely as the squares of the periodic times applied to the radii, that is, in the ratio compounded of the radii directly and the duplicate ratio of the periodic times inversely: $F \propto r/T^2$.

Corollary 3. So if the periodic times are equal, and therefore the velocities are as the radii, the forces are also as the radii, and conversely.

Corollary 4. If both the periodic times and the velocities are in the subduplicate ratio of the radii, the forces are equal, and conversely.

Corollary 5. If the periodic times are as the radii, and therefore the velocities are equal, the forces are inversely as the radii, and conversely.

Corollary 6. If the periodic times are in the sesquiplicate ratio of the radii, and therefore the velocities inversely in the subduplicate ratio of the radii, the forces are inversely as the squares of the radii, and conversely: $T \propto r^{3/2} \iff v \propto r^{-1/2} \iff F \propto r^{-2}$. This corrects the 1687 velocity clause.

Corollary 7. And universally, if the periodic time is as any power R^n of the radius R , and therefore the velocity inversely as R^{n-1} , the centripetal force is inversely as R^{2n-1} , and conversely.

Corollary 8. The old Corollary 7, with the addition that it is applied by substituting the uniform description of areas for uniform motion, and the distances of the bodies from the centres for the radii.

Corollary 9. From the same proof it also follows that the arc which a body describes in any time, revolving uniformly in a circle under a given centripetal

force, is a mean proportional between the diameter of the circle and the descent of the body falling under the same force in the same time: $s^2 = d \cdot h$.

1726. Corollaries 1 and 2 drop the “applied to the radii” formulation and keep only the compound ratios: in the ratio compounded of the duplicate ratio of the velocities directly and the simple ratio of the radii inversely; and in the ratio compounded of the radii directly and the duplicate ratio of the periodic times inversely.

Scholium

The case of Corollary 6 holds in the celestial bodies (as Wren, Hooke and Halley also gathered separately), and so what concerns a centripetal force decreasing in the duplicate ratio of the distances from the centres is set out more fully in what follows.

The preceding proof also gives the proportion of a centripetal force to any known force, such as gravity. In the time the body traverses the arc BC the force impels it through the space CD , which at the very beginning of the motion equals the square of the arc BD applied to the diameter of the circle; and every body urged by the same force continued always in the same direction describes spaces in the duplicate ratio of the times. So the force, in the time the revolving body describes any given arc, would make the same body, going straight on, describe a space equal to the square of that arc applied to the diameter of the circle; it is therefore to gravity as that space is to the space a heavy body describes falling in the same time. By propositions of this kind Huygens, in his excellent treatise *De Horologio oscillatorio*, compared the force of gravity with the centrifugal forces of revolving bodies.

In modern terms: $CD \approx s^2/d$ with $d = 2r$, so the force is to gravity as s^2/d is to the distance $\frac{1}{2}gt^2$ fallen in the same time.

1713. This paragraph is replaced. By the preceding Proposition and its corollaries one also obtains the proportion of a centripetal force to any known force, such as gravity: if a body revolves by its gravity in a circle concentric with the Earth, that gravity is its centripetal force, and from the descent of heavy bodies both the time of one revolution and the arc described in any given time are given, by Corollary 9. Huygens compared gravity with the centrifugal forces of revolving bodies by propositions of this kind.

The preceding can also be proved this way. In any circle inscribe a polygon of any number of sides. If a body moving along the sides with a given velocity is reflected from the circle at each vertex, the force with which it strikes the circle at each reflection is as its velocity; so the sum of the forces in a given time is as that velocity and the number of reflections jointly, that is (the polygon being given in kind) as the length described in that time and the same length applied to the radius, that is, as the square of that length applied to the radius; and so, if the polygon, its sides diminished indefinitely, coincides with the circle, as the square of the arc described in the given time applied to the radius. This is the force with which the body presses on the circle, and equal to it is the contrary force with which the circle continually repels the body toward the centre.

1713. “This is the *centrifugal force* with which the body presses on the

circle." **1726** replaces "the same length applied to the radius" by "the same length increased or diminished in the ratio of that length to the radius of the circle".

In modern terms: on a regular n -gon the body makes about $\nu t n/(2\pi r)$ reflections in the time t , each of impulse about $m\nu \cdot 2\pi/n$, so the total impulse is about $m\nu^2 t/r$. For a given t the length described is as ν , so this is as $(\nu t)^2/r$, Newton's square of the length applied to the radius.